Lesson 1: Multiplying and Factoring — The Reverse Relationship

Unwinding expansion into factors, reviewing the GCF, and checking every factoring.

1. What Is Factoring?

Expanding multiplies a factor across brackets: $4(x+3) = 4x+12$. **Factoring** runs the same road backwards — it rewrites an expanded expression as a **product of factors**: $4x+12 = 4(x+3)$. Likewise $2(x+5) = 2x+10$ unwinds to $2x+10 = 2(x+5)$, and $x(x+7) = x^2+7x$ unwinds to $x^2+7x = x(x+7)$. If multiplying your factors back out reproduces the original exactly, the factoring is correct.

Example 1

Each multiplication fact is done for you. Use it to write the matching factoring fact.

(a) Since $3(x+4) = 3x+12$, then $3x+12 =$ _____(_____)

(b) Since $8(y-1) = 8y-8$, then $8y-8 =$ _____(_____)

(c) Since $x(x+9) = x^2+9x$, then $x^2+9x =$ _____(_____)

(d) Since $4x(x-5) = 4x^2-20x$, then $4x^2-20x =$ _____(_____)

2. Recall: The GCF of Numbers

The **greatest common factor (GCF)** of two numbers is the largest number dividing evenly into both. Find it by **prime factorization**: break each number down, then multiply the primes appearing in *every* list. Since $12 = 2 \times 2 \times 3$ and $18 = 2 \times 3 \times 3$ share one 2 and one 3, the GCF is $2 \times 3 = 6$.

Example 2

Find the GCF of each pair. Show the prime factorization of each number first.

(a) 10, 15 (b) 8, 20 (c) 9, 24 (d) 16, 28

3. The GCF of a Monomial Set

The same idea works for **monomials** — just add the variable letters to the list of factors. Since $6x^3 = 2 \times 3 \times x \times x \times x$ and $9x^2 = 3 \times 3 \times x \times x$, the common factors are one 3 and two x's, so the GCF is $3x^2$. **Shortcut:** take the GCF of the coefficients, then each common variable raised to its *lowest* exponent.

Example 3

Find the GCF of each set of monomials, by prime factorization or by inspection.

(a) $4a^2,\ 10a$ (b) $12x^2y,\ 8xy^2$ (c) $15m^3n^2,\ 25m^2n$ (d) $9p^2q^2,\ 6pq^3$

4. Verifying a Factoring Is Correct

To check a factored form, **expand it back out**: if the product reproduces the original expression exactly, the factoring is correct. $3(x+3) = 3x+9$ ✓. But $4(x+3) = 4x+12$, which does *not* match $4x+8$ — since the GCF of $4x$ and 8 is 4, the correct form is $4(x+2)$.

Example 4

Expand each proposed factored form to check it. If it does not match, write the correct form.
(a) Is $4x + 12 = 4(x + 3)$ correct? (b) Is $5x + 15 = 5(x + 2)$ correct?

(c) Is $3x^2 + 6x = 3x(x + 2)$ correct? (d) Is $8x + 10 = 2(4x + 6)$ correct?

5. Putting It Together: Factor, Then Verify

The routine is *find the GCF, divide, write, verify.* For $6x+18$: the GCF is 6; $6x \div 6 = x$ and $18 \div 6 = 3$; so $6x+18 = 6(x+3)$; check by expanding, $6(x + 3) = 6x + 18$ ✓.

Example 5

Factor each expression completely, and verify your answer by expanding it back out.
(a) $4x + 8$ (b) $9x + 12$ (c) $6x^2 + 8x$

Lesson 2: Common Factors: Finding and Removing the GCF

Factoring out the greatest common factor, including variable and negative GCFs.

1. Recall the GCF, Then Remove It

The **GCF** of a set of monomials is the GCF of the coefficients times each common variable at its *lowest* exponent — so the GCF of $12x^2$ and $18x$ is $6x$. To **factor**, find the GCF, divide *every* term by it (a term equal to the GCF leaves a 1 behind — don't drop it), and write GCF(what's left). Always check by expanding: $10x + 15 = 5(2x + 3)$ because $5(2x + 3) = 10x + 15$.

Example 1

Factor each binomial by removing the GCF. Verify by expanding.
(a) $6x + 8$ (b) $12y - 9$ (c) $10a + 25$ (d) $14b - 21$

2. Factoring a Trinomial

The same routine works on three terms; there is simply one more term to divide in Step 2. For $6x^2 + 12x + 18$ the GCF is 6, and dividing each term gives $6(x^2 + 2x + 3)$.

Example 2

Factor each trinomial by removing the GCF. Verify by expanding.
(a) $3x^2 + 6x + 9$ (b) $10a^2 - 15a - 25$ (c) $8y^2 + 12y - 4$

3. A GCF That Includes a Variable

If *every* term shares a variable, that variable joins the GCF at its lowest exponent. For $8x^3 + 12x^2$: $\gcd(8, 12) = 4$ and the lowest power of x is x^2, so the GCF is $4x^2$ and $8x^3 + 12x^2 = 4x^2(2x + 3)$.

Example 3

Factor each polynomial by removing a GCF that includes a variable. Verify by expanding.
(a) $8x^3 + 12x^2$ (b) $12a^2b + 18ab^2$ (c) $4x^3 + 8x^2 + 12x$

4. Special Case: A Negative GCF

When the leading term is negative, it is customary to factor out a **negative** GCF so that what is left behind has a positive leading term. For $-4x - 12$, factor out -4 rather than 4: $-4x \div (-4) = x$ and $-12 \div (-4) = 3$, giving $-4(x + 3)$.

Example 4

Factor each polynomial by removing a *negative* GCF. Verify by expanding.
(a) $-4x - 12$ (b) $-5a + 15$ (c) $-3x^2 + 9x - 6$

5. Putting It Together: Negative GCF with a Variable

A GCF can carry a negative sign *and* a variable at once. For $-6x^3 - 15x^2$: $\gcd(6, 15) = 3$, the lowest power of x is x^2, and the leading term is negative, so the GCF is $-3x^2$ and the factored form is $-3x^2(2x + 5)$.

Example 5

Factor each polynomial completely. Verify by expanding.
(a) $-6x^3 - 15x^2$ (b) $-8a^3 + 12a^2 - 4a$ (c) $-10x^2y - 15xy^2$

Lesson 3: Factoring Trinomials of the Form $x^2 + bx + c$, Part 1: c Positive

Running FOIL backwards with the sum–product pattern, while the constant term stays positive.

1. From FOIL Back to Factors

FOIL turns $(x+2)(x+9)$ into $x^2+9x+2x+18 = x^2+11x+18$. **Factoring** runs that in reverse: starting from $x^2+11x+18$, find your way back to $(x + 2)(x + 9)$ — without testing every pair. The trick is to notice exactly where the 11 and the 18 came from.

Example 1

Expand each product. Compare the coefficient of x (that is b) and the constant (that is c) to the two numbers in the binomials.
(a) $(x + 2)(x + 6)$ (b) $(x + 4)(x + 5)$ (c) $(x + 1)(x + 8)$ (d) $(x + 3)(x + 3)$

What do you notice about the connection between b and the two numbers? Between c and the two numbers?

2. The Sum–Product Rule

In general $(x+p)(x+q) = x^2 + (p+q)x + pq$, so comparing with $x^2 + bx + c$ gives $b = p+q$ and $c = pq$. To factor $x^2 + bx + c$, find two numbers that **multiply to** c **and add to** b; those numbers are p and q, and $x^2 + bx + c = (x + p)(x + q)$.

Example 2

Complete the table by finding two **positive** integers with the given sum and the given product.

Sum	Product	Integers
7	10	
9	20	
10	24	
13	36	

3. Factoring When b and c Are Both Positive

If c is positive the two numbers share the same sign, and if b is also positive that sign must be positive — two positive numbers give both a positive product and a positive sum. So for $x^2 + 7x + 12$, look for two positive numbers with product 12 and sum 7.

Example 3

Factor each trinomial. In each case both numbers in your factors should be positive.

(a) $x^2 + 7x + 12$ (b) $x^2 + 9x + 18$ (c) $x^2 + 11x + 24$

4. Factoring When b Is Negative and c Is Positive

With c positive the two numbers still share a sign, but a negative b forces both to be **negative**. For $x^2 - 8x + 15$ we need product 15 and sum -8, giving -3 and -5, so $x^2 - 8x + 15 = (x - 3)(x - 5)$. **Rule of thumb:** c positive $\Rightarrow$ same sign, and that sign matches b.

Example 4

Factor each trinomial. In each case both numbers in your factors should be negative.

(a) $x^2 - 7x + 12$ (b) $x^2 - 9x + 18$ (c) $x^2 - 11x + 24$

5. Checking Your Answer, and Common Factors

Check any factoring by expanding it back with FOIL: if you recover the original trinomial, you are correct. If the terms share a **common factor**, remove it *first* — factoring is not finished until nothing further can be factored.

Example 5

Complete each of the following.

(a) It is claimed that $x^2 + 13x + 30 = (x + 3)(x + 10)$. Check by expanding with FOIL.

(b) Factor $x^2 + 13x + 36$.

(c) Factor completely: $3x^2 + 27x + 60$. (*Remove the common factor first.*)

Lesson 4: Factoring Trinomials $x^2 + bx + c$, Part 2: c Negative

Opposite signs, systematic factor-pair searches, prime trinomials, and removing a GCF first.

1. When c Is Negative: Opposite Signs

To factor $x^2 + bx + c = (x + p)(x + q)$ we still need two integers with **product** c and **sum** b. A *negative* product forces p and q to have **opposite signs**, and the integer with the **larger absolute value** takes the **same sign as** b — it controls the sum. So $x^2 + 3x - 18 = (x + 6)(x - 3)$, while $x^2 - 3x - 18 = (x - 6)(x + 3)$: same pair 6 and 3, signs swapped.

Example 1

Factor each trinomial. Both parts use the same two numbers, just with the signs swapped.
(a) $x^2 + 9x - 22$ (b) $x^2 - 9x - 22$

Example 2

Factor each trinomial.
(a) $x^2 + 2x - 24$ (b) $x^2 - 10x - 24$ (c) $n^2 + n - 30$ (d) $n^2 - n - 30$

2. A Systematic Approach for Trickier Cases

When c has many factor pairs, guessing is slow and easy to get wrong. Instead, list **every** factor pair of c in order starting from 1, and check the sum each pair gives in both sign arrangements until one matches b.

Example 3

Factor $x^2 + 5x - 36$ and $x^2 - 5x - 36$. First complete the table of factor pairs of 36.

Factor Pair	**Product**	**Sum (larger +)**	**Sum (larger −)**
1, 36	−36		
2, 18	−36		
3, 12	−36		
4, 9	−36		
6, 6	−36		

(a) Use the row with a sum of 5 to factor $x^2 + 5x - 36$. (b) Use the row with a sum of -5 to factor $x^2 - 5x - 36$.

3. Prime Trinomials

If you list **every** factor pair of c and none of them — in either sign arrangement — sums to b, the trinomial cannot be factored over the integers; we call it **prime**. Check every pair before concluding this, since stopping too early is the most common mistake.

Example 4

Decide whether each can be factored over the integers. If it can, factor it; if not, list the factor pairs of c you checked and explain why none work.
(a) $x^2 + 4x - 15$ (b) $x^2 + 4x - 12$

4. Removing a Common Factor First

Always look for a **greatest common factor** among all the terms *before* factoring the trinomial; removing it leaves smaller numbers to work with. Factoring is not finished until nothing more can be pulled out.

Example 5

Factor each expression completely.
(a) $2x^2 + 4x - 30$ (b) $3x^2 - 6x - 45$ (c) $2x^3 + 2x^2 - 24x$

Lesson 5: Factoring the Difference of Squares

Running the sum-and-difference shortcut in reverse, and knowing when it does not apply.

1. Reversing the Pattern

Multiplying a sum by a difference cancels the middle terms: $(a+b)(a-b) = a^2 - b^2$. Read that backwards and you have a factoring rule: $a^2 - b^2 = (a+b)(a-b)$. Take the square root of each term, then write one binomial with a $+$ and one with a $-$: $x^2 - 64 = x^2 - 8^2 = (x+8)(x-8)$.

Example 1

Factor each difference of squares.
(a) $q^2 - 1$ (b) $m^2 - 16$ (c) $c^2 - 100$ (d) $x^2 - y^2$

2. Identifying Perfect Squares Quickly

Know the small squares on sight: $1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225$. A variable power is a perfect square when its exponent is **even** ($x^4 = (x^2)^2$, $x^6 = (x^3)^2$), and a monomial is a perfect square when its coefficient is a square *and* every exponent is even — so $\sqrt{4x^2} = 2x$, $\sqrt{16x^4} = 4x^2$, $\sqrt{25y^6} = 5y^3$.

Example 2

Factor each difference of squares.
(a) $4x^2 - 9$ (b) $9a^2 - 1$ (c) $x^4 - 16$ (d) $25p^2 - 4q^2$

3. Remove the GCF First

Always check for a greatest common factor before anything else. A binomial may not look like a difference of squares until the GCF is out: neither $2x^2$ nor 18 is a perfect square, but $2x^2 - 18 = 2(x^2 - 9) = 2(x-3)(x+3)$. Factor *completely* — keep going until nothing left inside the brackets can be factored.

Example 3

Factor completely by first removing the greatest common factor.
(a) $2x^2 - 18$ (b) $3m^2 - 48$

Example 4

Factor completely by first removing the greatest common factor.
(a) $5x^3 - 45x$ (b) $8a^2b^2 - 32$

4. When It Is NOT a Difference of Squares

The pattern needs *both* conditions: subtraction, and two perfect squares. A **sum** of squares such as $x^2 + 9$ cannot be factored over the integers. Neither can $x^2 - 20$, because 20 is not a perfect square (16 and 25 are its neighbours). Check both before you factor.

Example 5

Decide whether each is a difference of squares. If it is, factor it; if not, explain why not.
(a) $x^2 + 9$ (b) $x^2 - 9$ (c) $4y^2 - 15$ (d) $16x^2 - 1$

Lesson 6: Factoring Perfect Square Trinomials

Reversing the squared-binomial pattern, and telling a true perfect square from an impostor.

1. The Pattern and the Three-Part Test

Squaring a binomial always gives three terms: $a^2 + 2ab + b^2 = (a + b)^2$ and $a^2 - 2ab + b^2 = (a - b)^2$. Read backwards, a trinomial is a **perfect square trinomial** only if it passes the **three-part test**: the first term is a perfect square, the last term is a perfect square, and the middle term equals $2 \times \sqrt{\text{first}} \times \sqrt{\text{last}}$. For $t^2 + 16t + 64$: roots t and 8, and $2(t)(8) = 16t$ matches — all three pass.

Example 1

Use the three-part test to decide whether each trinomial is a perfect square trinomial.
(a) $x^2 + 6x + 9$ (b) $x^2 - 6x + 9$ (c) $x^2 + 7x + 10$ (d) $x^2 + 10x - 25$

2. Factoring the Sum Case

If the test passes and the middle term is **positive**, take the square root of the first term and of the last term, then write their *sum*, squared. For $x^2 + 10x + 25$: $\sqrt{x^2} = x$, $\sqrt{25} = 5$, and $2(x)(5) = 10x$ ✓, so $x^2 + 10x + 25 = (x + 5)^2$. No guess-and-check is needed.

Example 2

Factor each perfect square trinomial.
(a) $x^2 + 8x + 16$ (b) $x^2 + 14x + 49$ (c) $x^2 + 20x + 100$ (d) $x^2 + 18x + 81$

3. Factoring the Difference Case

Now compare $x^2 - 10x + 25$: same first and last terms, same roots x and 5, and $2(x)(5) = 10x$ still matches — but the middle term is negative, so the binomial being squared is a **difference**: $x^2 - 10x + 25 = (x - 5)^2$. Only the middle sign changes. The last term is always added, since a square is never negative.

Example 3

Factor each perfect square trinomial.
(a) $x^2 - 8x + 16$ (b) $x^2 - 14x + 49$ (c) $x^2 - 20x + 100$ (d) $x^2 - 18x + 81$

4. Perfect Square, or an Ordinary Trinomial?

Most trinomials are *not* perfect squares, so run the test first. If it passes, use the shortcut; if it fails — and the classic near-miss is right squares but a wrong middle term, like $x^2 + 10x + 24$ — fall back on the **sum-product method**: two numbers that multiply to c and add to b.

Example 4

Decide whether each is a perfect square trinomial, then factor completely by the appropriate method.
(a) $x^2 + 9x + 14$ (b) $x^2 + 14x + 49$ (c) $x^2 - 11x + 18$ (d) $x^2 - 16x + 64$

Example 5

Decide whether each is a perfect square trinomial, then factor completely by the appropriate method.
(a) $4x^2 + 12x + 9$ (b) $x^2 + 9x + 18$ (c) $9x^2 - 30x + 25$ (d) $x^2 - 5x - 14$

Lesson 7: Factoring $ax^2 + bx + c$ (where $a \neq 1$)

Decomposition: split the middle term, group in pairs, and factor out the common binomial.

1. Why the Sum-Product Method Is Not Enough

Expanding $(x+p)(x+q)$ always gives $x^2+(p+q)x+pq$ — the leading coefficient is always 1. So a trinomial like $2x^2+7x+3$ cannot come from two binomials that both start with a plain x. When $a \neq 1$ we need a method that accounts for the leading coefficient: **decomposition**.

2. The Decomposition Method, Step by Step

Step 1: multiply $a \times c$. **Step 2:** find two integers with product $a \times c$ and sum b. **Step 3:** split the middle term bx into those two terms (now four terms). **Step 4:** group into two pairs and factor the GCF out of each pair. **Step 5:** both pairs share a common binomial — factor it out. **Step 6:** check by expanding with FOIL.

Worked model — factor $2x^2 + 7x + 3$. Here $a = 2$, $b = 7$, $c = 3$, so $ac = 6$; the two integers with product 6 and sum 7 are 6 and 1. Split, group, and factor:

$$2x^2 + 7x + 3 = \underbrace{2x^2 + 6x}_{\text{GCF}=2x} + \underbrace{x + 3}_{\text{GCF}=1} = 2x(x+3) + 1(x+3) = (x+3)(2x+1).$$

Check: $(x+3)(2x+1) = 2x^2 + x + 6x + 3 = 2x^2 + 7x + 3$. ✓

Example 1

Factor using every step: find ac, find the two integers, split, group, factor out the binomial, then check with FOIL.
(a) $3x^2 + 10x + 3$ (b) $2x^2 + 9x + 4$

3. Practicing Decomposition

Once you know the six steps, the method is identical every time — only the numbers change. Here is more practice with a, b, and c all positive.

Example 2

Factor each trinomial using decomposition.
(a) $2x^2 + 11x + 5$ (b) $3x^2 + 13x + 4$ (c) $4x^2 + 12x + 5$

4. Decomposition with Negative b and/or c

Nothing changes except the **signs** of the two integers found in Step 2. If ac is positive the integers share a sign (both negative when b is negative); if ac is negative the integers have opposite signs, and the one with the larger absolute value takes the sign of b.

Example 3

Factor each trinomial. Here ac is positive, so both integers have the same sign.
(a) $2x^2 - 9x + 4$ (b) $3x^2 - 11x + 6$

Example 4

Factor each trinomial. Here ac is negative, so the two integers have opposite signs.
(a) $2x^2 + 7x - 4$ (b) $3x^2 - 4x - 4$ (c) $4x^2 - 4x - 3$

5. Removing a GCF First, Then Decomposing

Always check for a GCF *before* decomposing — it shrinks a, b, and c, so Steps 1 and 2 get much easier. For example, $4x^2 + 10x - 6 = 2(2x^2 + 5x - 3)$; then $ac = -6$ with sum 5 gives 6 and -1, so $2x^2 + 5x - 3 = 2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (x + 3)(2x - 1)$, and $4x^2 + 10x - 6 = 2(x + 3)(2x - 1)$.

Example 5

Factor each expression completely. Remove the GCF first, then decompose what remains.
(a) $6x^2 + 3x - 30$ (b) $4x^3 - 2x^2 - 30x$

Lesson 8: Choosing a Factoring Strategy

No new technique — a checklist for deciding which factoring method a new expression needs.

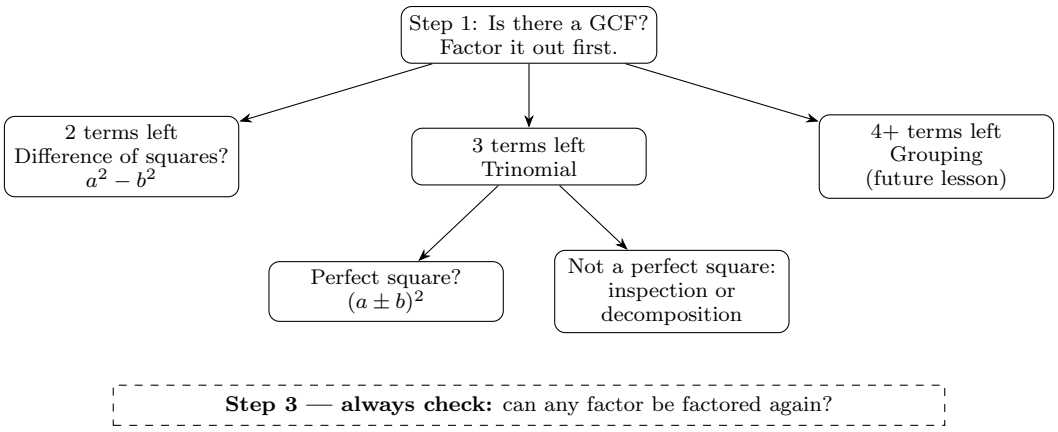
1. The Five Methods, Side by Side

You already know how to run each method; the new skill is picking the right one. Everything from Lessons 1–7 fits in this one table.

Method	Terms	What to look for	Pattern
GCF	any	Every term shares a common factor	$ab + ac = a(b + c)$
Difference of squares	2	Two perfect squares with a subtraction between them	$a^2 - b^2 = (a - b)(a + b)$
Perfect square trinomial	3	First and last terms are perfect squares; middle is 2×(their roots)	$a^2 \pm 2ab + b^2 = (a \pm b)^2$
Trinomial, $a = 1$	3	Leading coefficient is 1	Two numbers: product c, sum b
Trinomial, $a \neq 1$	3	Leading coefficient is not 1	Two numbers: product ac, sum b (decomposition)

2. The Factoring Strategy Checklist

Work these steps in order, every single time: **(1)** factor out the GCF first — this is the most commonly missed step; **(2)** count the terms that are left — 2 terms means check for a difference of squares, 3 terms means check for a perfect square trinomial first and otherwise factor by inspection ($a = 1$) or decomposition ($a \neq 1$), and 4 or more terms needs grouping (a future lesson); **(3)** check whether any factor can still be factored again — stopping one step early is the other common mistake.



Example 1

Name the method (or methods, in order) you would use. Do **not** factor yet.

(a) $x^2 - 4$ (b) $x^2 + 7x + 12$ (c) $4x^2 - 8x$

(d) $3x^2 + 13x + 4$ (e) $x^2 + 10x + 25$ (f) $2x^2 - 50$

3. The Checklist in Action: GCF, Then a Trinomial

Factor $2x^2 + 20x + 42$. **Step 1:** every term is divisible by 2, so $2x^2 + 20x + 42 = 2(x^2 + 10x + 21)$. **Step 2:** three terms remain; 21 is not a perfect square, so this is not a perfect square trinomial, and the leading coefficient is 1, so find two numbers with product 21 and sum 10 — namely 7 and 3 — giving $2(x + 7)(x + 3)$. **Step 3:** neither binomial factors again, so we are finished.

Example 2

Factor completely.

(a) $2x^2 + 14x + 20$ (b) $4x^2 - 8x - 60$

4. Two-Step Factoring: GCF, Then a Difference of Squares

When a two-term expression is not a difference of squares as written, remove the GCF and look again: $4x^2 - 64 = 4(x^2 - 16) = 4(x - 4)(x + 4)$. Leaving the answer as $4(x^2 - 16)$ is stopping one step too early.

Example 3

Factor completely. (Hint: this will take two steps.)

(a) $4x^2 - 64$ (b) $6x^2 - 24$

5. Mixed Practice

With no hint to tell you which method applies, run the checklist from the top: GCF, then count terms, then check the factors. Confirm any answer by expanding it back.

Example 4

Factor completely.
(a) $x^2 - 64$ (b) $x^2 - 3x - 40$

Example 5

Factor completely.
(a) $3x^2 + 11x + 6$ (b) $x^2 - 12x + 36$

Lesson 9: Solving Polynomial Equations by Factoring

Using the zero product property to turn one factored equation into two easy ones.

1. The Zero Product Property

Since $7 \times m = 0$ forces $m = 0$, a product can only be zero when one of its factors is zero: **if** $A \cdot B = 0$**, then** $A = 0$ **or** $B = 0$. So $(x-3)(x+5) = 0$ splits into $x - 3 = 0$ or $x + 5 = 0$, giving **two** solutions (roots), $x = 3$ and $x = -5$. A nonzero constant out front never produces a solution — only the factors containing x matter.

Example 1

Solve each equation using the Zero Product Property.
(a) $(x-4)(x+7) = 0$ (b) $(x+2)(x-9) = 0$ (c) $(x+6)(x+1) = 0$ (d) $4(x-5)(x+3) = 0$

2. Solving Quadratic Equations by Factoring

Most equations do not arrive factored. The process: make one side exactly 0, factor the other side completely (GCF first, then difference of squares or trinomial), set *each* factor equal to 0, and solve each small equation. Every equation below is already equal to 0, so factor right away.

Example 2

Solve each equation by factoring.
(a) $x^2 - 49 = 0$ (b) $x^2 + 9x = 0$

(c) $x^2 - 5x + 6 = 0$ (d) $x^2 + 2x - 24 = 0$

3. Equations That Need Rearranging First

The zero product property works only when the product equals 0 — you cannot split $x(x+3) = 10$ into $x = 10$ and $x+3 = 10$. Move every term to one side first: $x^2 + 5x = 14$ becomes $x^2 + 5x - 14 = 0$, which factors as $(x+7)(x-2) = 0$, so $x = -7$ or $x = 2$.

Example 3

Solve each equation. Rearrange first so that one side is 0.
(a) $x^2 + 3x = 10$ (b) $x^2 = 8x - 15$ (c) $x^2 - 4x = 12$

4. Checking Solutions by Substitution

Substitute each root back into the **original** equation, not the factored form. For $x^2+5x = 14$: $(-7)^2+5(-7) = 49-35 = 14$ ✓ and $(2)^2 + 5(2) = 4 + 10 = 14$ ✓, so both roots are confirmed.

Example 4

Solve $x^2 - x = 12$ by factoring, then check both solutions in the original equation.
Solve: Check:

5. Combining Skills: A Leading Coefficient Other Than One

When the leading coefficient is not 1, factor by **decomposition** first, then apply the zero product property exactly as before. Solutions may be fractions, which is perfectly normal.

Example 5

Solve $2x^2 + 5x = 3$ by factoring, then check both solutions in the original equation.
Solve: Check:

Lesson 10: Unit Review: Factoring

Every factoring type from the unit in one place, with a worked example for each.

Type	**Looks Like**	**Factored Form**
Common factor (GCF)	every term shares a factor	GCF × (what is left)
Trinomial, $a = 1$	$x^2 + bx + c$	$(x + m)(x + n)$ with $mn = c$, $m + n = b$
Difference of squares	$a^2 - b^2$	$(a + b)(a - b)$
Sum of squares	$a^2 + b^2$	cannot be factored
Perfect square trinomial	$a^2 \pm 2ab + b^2$	$(a \pm b)^2$
Trinomial, $a \neq 1$	$ax^2 + bx + c$	decompose: product ac, sum b, then group
Zero Product Property	$A \times B = 0$	$A = 0$ or $B = 0$

1. Removing the Greatest Common Factor

Find the GCF of the coefficients times the *lowest* power of each common variable, divide every term by it, and write GCF × (what's left). If the leading term is negative, factor out a **negative** GCF.

Example 1

Factor completely.
(a) $16x^2 - 24x$ (b) $-18a^3 + 27a^2 - 9a$

2. Trinomials $x^2 + bx + c$, c Positive

When c is positive the two integers (product c, sum b) have the **same** sign — both positive if b is positive, both negative if b is negative.

Example 2

Factor each trinomial.
(a) $x^2 + 15x + 44$ (b) $x^2 - 12x + 35$

3. Trinomials $x^2 + bx + c$, c Negative

When c is negative the two integers have **opposite** signs, and the one with the larger absolute value takes the same sign as b.

Example 3

Factor each trinomial.
(a) $x^2 + 2x - 35$ (b) $x^2 - x - 56$

4. Difference of Squares

Two perfect squares separated by subtraction, $a^2 - b^2$, factors instantly as $(a+b)(a-b)$. A *sum* of squares never factors this way.

Example 4

Factor each binomial.
(a) $x^2 - 121$ (b) $49x^2 - 25$

5. Perfect Square Trinomials

First and last terms are perfect squares and the middle term is *twice* the product of their roots: $a^2 + 2ab + b^2 = (a+b)^2$ and $a^2 - 2ab + b^2 = (a-b)^2$.

Example 5

Factor each perfect square trinomial.
(a) $x^2 + 28x + 196$ (b) $4x^2 - 20x + 25$

6. Factoring $ax^2 + bx + c$ by Decomposition

When $a \neq 1$: multiply $a \times c$, find two integers with that product that add to b, split the middle term with them, then factor by grouping.

Example 6

Factor by decomposition.
(a) $2x^2 + 13x + 6$ (b) $3x^2 - x - 10$

7. Choosing a Strategy

Step 1: always remove a GCF first. **Step 2:** count the terms left — **two terms**, check difference of squares; **three terms**, check perfect square, else sum–product ($a = 1$) or decomposition ($a \neq 1$). **Step 3:** check every factor for further factoring and verify by expanding.

Example 7

Factor each expression completely.
(a) $5x^2 - 80$ (b) $2x^2 + 16x + 32$ (c) $4x^3 - 4x^2 - 24x$

8. Solving Equations by Factoring

Move everything to one side so the equation equals 0, factor, then set each factor equal to 0 and solve (Zero Product Property).

Example 8

Solve each equation by factoring.
(a) $x^2 + x - 42 = 0$ (b) $3x^2 + 5x - 2 = 0$

9. Mixed Review: Putting It All Together

Each question below combines more than one skill from the unit — exactly the kind of question you might see on the unit test.

Challenge 1

Factor completely: $6x^3 + 3x^2 - 45x$.

Challenge 2

Solve: $3x^2 = 12x$.

Challenge 3

A rectangular garden has area $(x^2 + 3x - 54)$ m^2. One side length of the garden is $(x + 9)$ m.
(a) Determine a simplified expression for the other side length. (b) Determine the actual dimensions of the garden if $x = 8$.